

ECE 647: Lecture 12

1 Vector Function Decomposition

$$f(x) = h(g(x)) : g = [g_1 \dots g_k]^T$$

Assume **wlog** that $x \in \mathbb{R}$, restricting the function to a line.

$$h : \mathbb{R}^1 \rightarrow \mathbb{R}, h(y_1, \dots, y_k)$$

$$g_i : \mathbb{R} \rightarrow \mathbb{R}, g = [g_1 \dots g_k]^T$$

$$f(x) = h(g_1(x), g_2(x), \dots, g_k(x))$$

Then

$$f'(x) = \nabla h(g(x))^T \cdot [g'(x)] \in \mathbb{R}$$

$$f''(x) = [g'(x)]^T \nabla^2 h(g(x)) \cdot [g'(x)] + \nabla h(g(x))^T [g''(x)] \in \mathbb{R}$$

where:

$$\nabla h = \begin{bmatrix} \frac{\partial h}{\partial y_1} \\ \vdots \\ \frac{\partial h}{\partial y_k} \end{bmatrix}, \quad g' = \begin{bmatrix} g'_1 \\ \vdots \\ g'_k \end{bmatrix}, \quad g'' = \begin{bmatrix} g''_1 \\ \vdots \\ g''_k \end{bmatrix}, \quad \nabla^2 h = \left[\frac{\partial^2 h}{\partial y_i \partial y_j} \right]_{k \times k}$$

Positive or negative semi-definite.

Assume g_i is convex, $g''_i(x) \geq 0$ If $\nabla^2 h(\cdot) \geq 0$ then

$$g'(x)^T \nabla^2 h(g(x)) g'(x) \geq 0$$

If in addition, $\nabla h(f(x)) \geq 0$ for each element, then

$$\nabla h(f(x))^T g''(x) \geq 0$$

We then obtain the following rule

- The convexity of f must follow that of h .
- If h and g are of the same type, need h increasing in every argument.
- If h and g are of different types, need h decreasing in every argument.

Finally, if $\text{dom } h \neq \mathbb{R}^k$, we will need to replace h by an extended-value function $\bar{h}$.

Example:

(a) $h(\mathbf{z}) = z_{(1)} + \dots + z_{(r)}$, the sum of the r largest components.

- $-h(\mathbf{z})$ is convex.
- $g_1(x), \dots, g_k(x)$ are convex.

$\Rightarrow h(g_1(x), \dots, g_k(x))$ is convex because $h(\mathbf{z})$ is non-decreasing in each argument.

(b) $h(\mathbf{z}) = \log \left(\sum_{i=1}^k e^{-z_i} \right)$

- $g_1(x), \dots, g_k(x)$ are convex.

$\Rightarrow h(g_1, \dots, g_k) = \log \left(\sum_{i=1}^k e^{-g_i(x)} \right)$ is convex.

(c) $f(x, u, v) = \sqrt{uv - x^T x}, \quad x \in \mathbb{R}^n, u, v \in \mathbb{R}$

on $u > 0, v > 0, uv > x^T x$.

See Boyd Ex. 3.22(b), p. 117.

$$f(x, u, v) = \sqrt{u \left(v - \frac{x^T x}{u} \right)} = \sqrt{\delta_1 \cdot \delta_2}$$

where $\delta_1(x, u, v) = u, \quad \delta_2(x, u, v) = v - \frac{x^T x}{u}$

- $\sqrt{\delta_1 \delta_2}$ is concave in δ_1, δ_2 .
- $\delta_2(x, u, v)$ is concave in (x, u, v) .
- $\delta_1(u) = u$ is concave.
- $\sqrt{\delta_1 \delta_2}$, extended to $\mathbb{R}^2$, is non-decreasing in δ_1 and δ_2 .

2 Summary

- Convex functions

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

- First order condition

$$f(y) \geq f(x) + f'(x)(y - x)$$

- Second order condition

$$f''(x) \geq 0$$

- Operations

- Non-negative weighted sum
- Affine change of variable
- Pointwise maximum

- Pointwise minimum (*more demanding*)
- Composition
- Perspective

3 Jensen's Inequality

If a function f is convex, then

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

for any $0 \leq \theta \leq 1$.

Easily extended to finite convex combinations:

$$f(\theta_1 x_1 + \cdots + \theta_k x_k) \leq \theta_1 f(x_1) + \cdots + \theta_k f(x_k)$$

for all $\theta_1 + \cdots + \theta_k = 1$, $0 \leq \theta_1, \dots, \theta_k \leq 1$.

Can also be extended to infinite convex combinations:

Let $\theta_1, \theta_2, \dots \geq 0$ satisfy

$$\sum_{k=1}^{\infty} \theta_k = 1$$

Let $x_1, x_2, \dots \in \text{dom } f$, which is convex, and the limit

$$\sum_{k=1}^{\infty} \theta_k x_k$$

exists.

Then:

1. $\sum_{k=1}^{\infty} \theta_k x_k \in \text{dom } f$

$$2. f(\sum_{k=1}^{\infty} \theta_k x_k) \leq \sum_{k=1}^{\infty} \theta_k f(x_k)$$

The proof of part 1 will need the Separation Theorem (to be discussed later).

The proof of part 2 uses the first-order condition (assuming f is differentiable) or we use subgradients (to be discussed later).

We can also extend to integrals:

$p(x) \geq 0$ on $S \subseteq \text{dom } f$ — "density"

$$\int_S p(x) dx = 1$$

$$\Rightarrow f\left(\int_S p(x) \cdot x dx\right) \leq \int_S f(x)p(x) dx$$

if f is convex.

Same for expectations:

$$f(\mathbb{E}[\bar{X}]) \leq \mathbb{E}[f(\bar{X})]$$

4 Convex Optimization Problems

- A convex optimization problem minimizes a convex function over a convex set:

$$- \min f_0(x) \quad (\text{convex function})$$

$$- \text{s.t. } x \in C \quad (\text{convex set})$$

However, for the most part, we will use the following *standard form*:

- **Standard form:**

$$\min f_0(x)$$

$$\text{s.t. } f_i(x) \leq 0, \quad i = 1, \dots, m$$

$$h_i(x) = 0, \quad i = 1, \dots, p$$

where f_0, f_i are convex functions and h_i are linear functions.

- x : variable to optimize

$f_0(x)$: objective

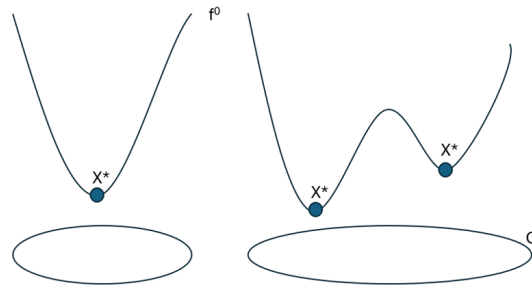
$f_i(x) \leq 0$: inequality constraints

$h_i(x) = 0$: equality constraints

- Domain of the problem:

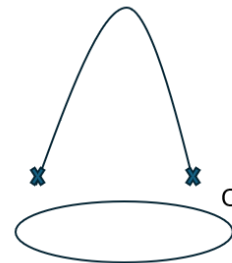
$$D = \bigcap_{i=0}^m \text{dom } f_i \cap \bigcap_{i=1}^p \text{dom } h_i$$

- A key feature is that any local optimum must be a global optimum.



- What about concave functions?

- min a concave function won't work.



- max a concave function f_0 is the same as min the convex function $-f_0$.

- $f_i(x) \geq 0$ for a concave function is the same as $-f_i(x) \leq 0$ for the convex function $-f_i(x)$.

- Why $h_i(x)$ must be linear?

- Some properties of convex optimization problems hold for the *general form*, but some require the *stricter form*.

- A point x is *feasible* if $x \in D$ and x satisfies the constraints.

- Optimal value:

$$p^* = \inf \{f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m, \quad h_i(x) = 0, i = 1, \dots, p\}$$

- We allow:

- $p^* = +\infty$ (if no point x is feasible).

- $p^* = -\infty$ (if there exists a sequence of feasible x_k , such that $f(x_k) \rightarrow -\infty$).

- Optimal point:

If the optimal value p^* is attained at feasible point x^* ,

$\Rightarrow x^*$ is an optimal point.

- We have taken the inequality constraints as " $\leq$ ". Although some results also apply for " $<$ ".

The " $\leq$ " part often implies that the feasible set is closed. Hence, the optimal value is always attainable if $f(x)$ is continuous.